FUNDAMENTAL CONCEPTS OF IEEE 802.3 ETHERNET ARE DISCUSSED

CSMA IS EXPLAINED

IEEE ADDRESS FORMAT IS PRESENTED

1. Multiple Access
2. CSMA/CD
3. IEEE 802.3 CSMA/CD - Ethernet
4. IEEE 802 Address Format
5. Address Resolution Protocol

Note: This lecture is based on Chapter 5 of the textbook (Kurose and Ross) and the figures are provided by the authors.

Multiple Access

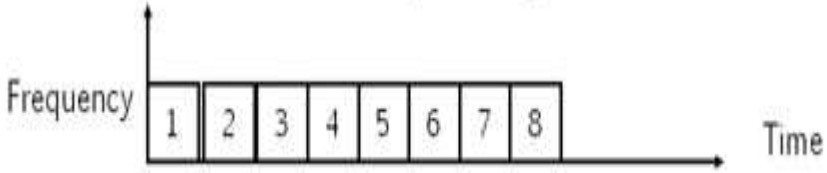
(a) Aloha Multiple Access

(b) Carrier-Sense Multiple Access with Collision Detection

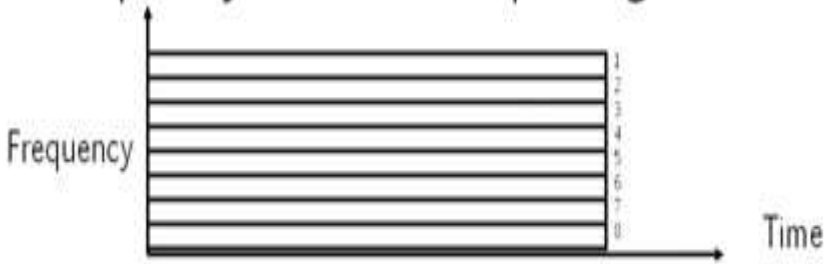
Multiple Access

› How multiple users can share a link?

› Time Division Multiplexing



› Frequency Division Multiplexing



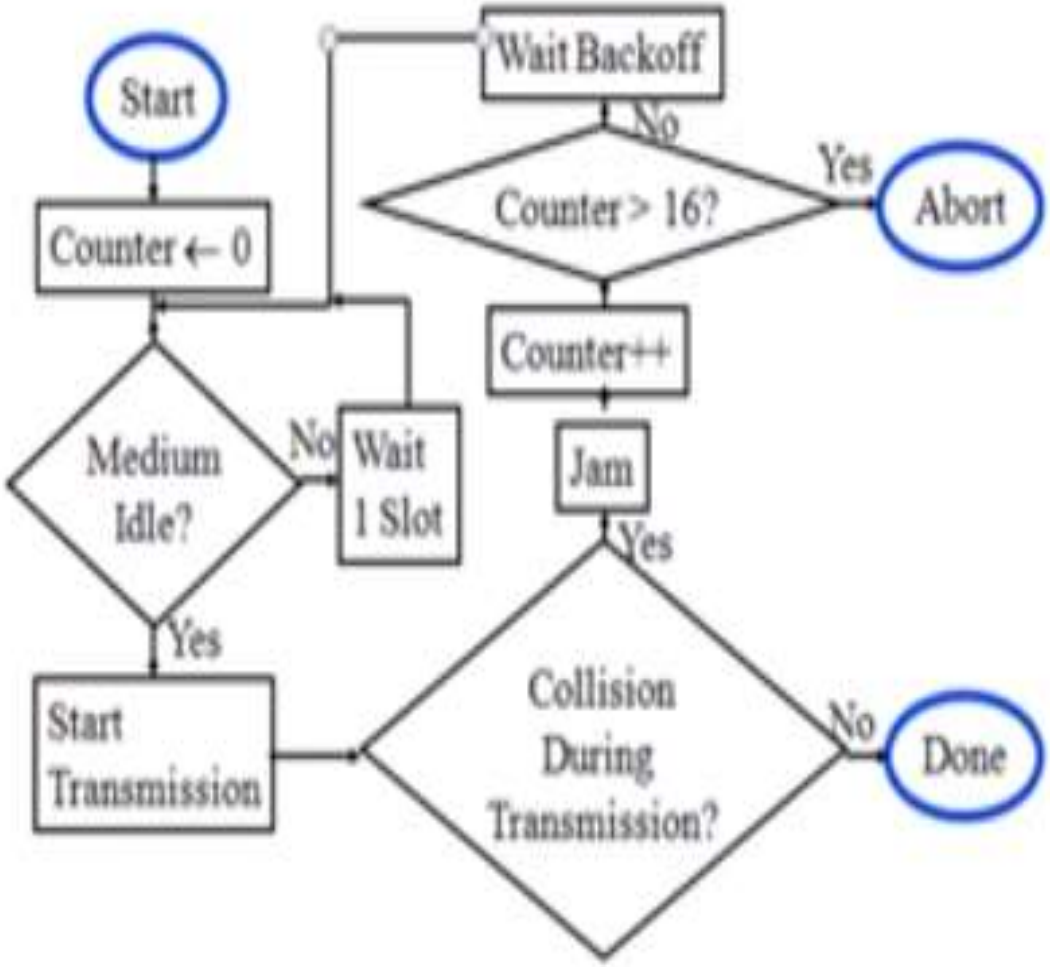
CSMA/CD

- Aloha at Univ of Hawaii:
  Transmit whenever you like
  Worst case utilization = 1/(2e) =18%
- Slotted Aloha: Fixed size transmission slots
  Worst case utilization = 1/e = 37%
- CSMA: Carrier Sense Multiple Access
  Listen before you transmit
- p-Persistent CSMA: If idle, transmit with probability p. Delay by one time unit with probability 1-p
- CSMA/CD: CSMA with Collision Detection
  Listen while transmitting. Stop if you hear someone else

IEEE 802.3 CSMA/CD

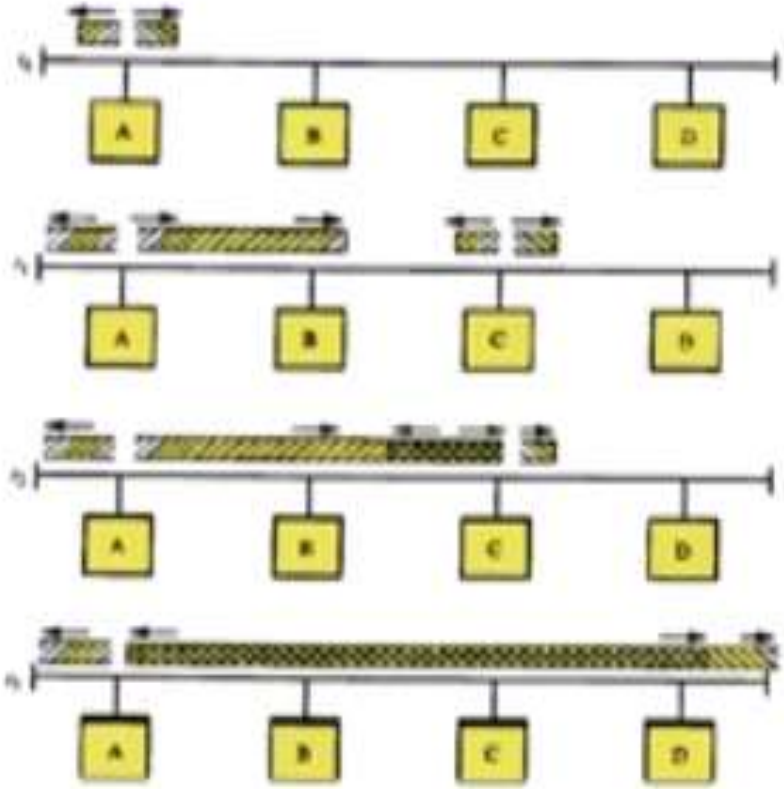
- If the medium is idle, transmit (1-persistent).
- If the medium is busy, wait until idle and then transmit immediately.
- If a collision is detected while transmitting,
  - Transmit a jam signal for one slot (= 51.2 μs = 64 byte times)
  - Wait for a random time and reattempt (up to 16 times)
  - Random time = Uniform$[0, 2^{\min(k,10)}-1]$ slots
- Collision detected by monitoring the voltage
  High voltage ⇒ two or more transmitters ⇒ Collision
  ⇒ Length of the cable is limited to 2 km

IEEE 802.3 CSMA/CD Flow Chart



CSMA/CD Operation

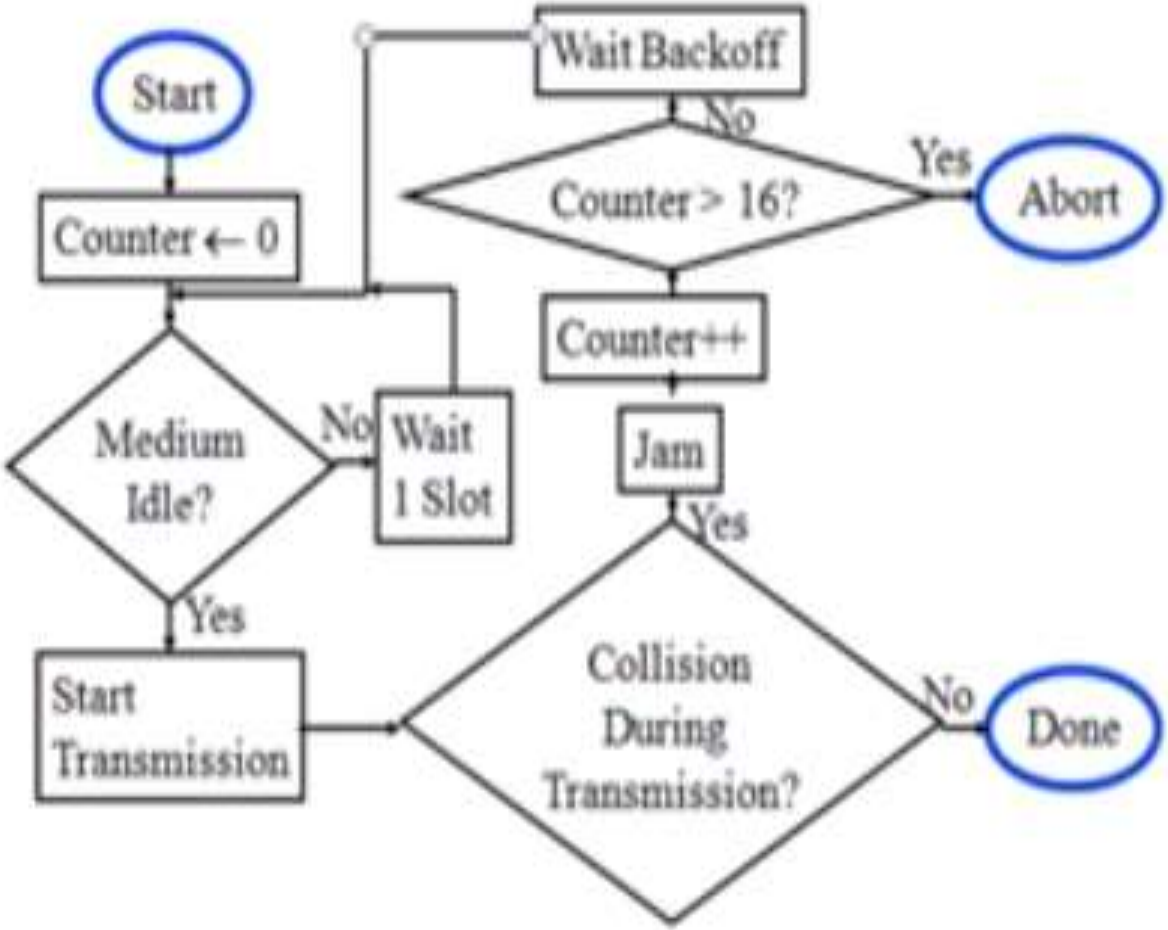
› Collision window = 2 × One-way Propagation delay = 51.2 μs



One way delay = 25.6 μs

Max Distance <2.5 km

IEEE 802.3 CSMA/CD Flow Chart



IEEE 802 Address Format

- 48-bit:1000 0000 : 0000 0001 : 0100 0011 : 0000 0000 : 1000 0000 : 0000 1100 = 80:01:43:00:80:0C

Organizationally Unique Identifier (OUI)			24 bits assigned by OUI Owner
Individual/ Group	Universal/ Local		
1	1	22	24

- Multicast = "To all bridges on this LAN"
- Broadcast = "To all stations" = 111111....111 = FF:FF:FF:FF:FF:FF